SHARP CURVATURE AND FINITE-BOX CRITERIA FOR LATTICE YANG–MILLS THEORY

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Abstract. The Riemannian Hessian of the Wilson action on d-dimensional lattice gauge fields with values in $\mathrm{SU}(N)$, $\mathrm{SO}(N)$ or $\mathrm{U}(N)$ is bounded by $4d$, and the bound is attained. Consequently the lattice Yang–Mills measure satisfies a log-Sobolev inequality uniform in the volume for $|\beta| < 1/(8d)$, no uniform curvature argument reaches further, and the SU(2) Gibbs state is unique for $|\beta| < 1/(8(d-1))$. We also prove a gauge-invariant finite-box criterion for decay of correlations, and we isolate what is still missing for a mass gap at intermediate coupling.

1. Introduction

Wilson's lattice gauge theory [26] with compact structure group G is the standard starting point for a construction of four-dimensional Yang–Mills theory. At strong coupling, that is for small inverse coupling β, the theory is under control: the Osterwalder–Seiler cluster expansion [22, 23] gives a transfer-matrix gap and exponential clustering, and Shen, Zhu and Zhu [25] obtained a log-Sobolev inequality, uniform in the volume, by verifying the Bakry–Émery criterion [1, 2] on the product of copies of G. Their argument rests on a bound

$$|\operatorname{Hess} S_0(v,v)| \le 8(d-1)\,|v|^2 \tag{1}$$

for the Hessian of the Wilson action $S_0 = \sum_p \operatorname{Re}\operatorname{Tr} Q_p$, and gives the regime $|\beta| < 1/(16(d-1))$ in their normalisation. The same constant $8(d-1)$ is used in later work on the Langevin dynamics, the Yang–Mills–Higgs model and the 't Hooft regime [9, 21, 24].

The first purpose of this paper is to determine the best constant in (1). The cross terms between the four links of one plaquette, which are bounded plaquette by plaquette in (1), reassemble over the whole lattice into the square of a covariant divergence. This yields the following.

Theorem A (Sharp Hessian bound)**.** *Let $G \in \{\mathrm{SU}(N), \mathrm{SO}(N), \mathrm{U}(N)\}$ and $d \ge 2$. For every gauge field Q on $\mathbb{Z}^d$ or on a torus and every finitely supported tangent vector $v = (X_e Q_e)_e$,*

$$\pm \operatorname{Hess} S_0(v,v) + \sum_x |D_x X|^2 \le 4d\,|v|^2, \qquad \textit{in particular} \qquad |\operatorname{Hess} S_0(v,v)| \le 4d\,|v|^2,$$

where $D_x X$ is the covariant divergence (6)*. For every $N \ge 2$ and $d \ge 2$ the constant $4d$ is attained with both signs.*

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For $d = 2$ the constants $4d$ and $8(d-1)$ coincide; for $d = 3$ and $d = 4$ they are 12 against 16 and 16 against 24. Within the literature we searched (arXiv and journals to October 2026) no source states the bound $4d$.

Theorem B (Strong-coupling regime)**.** *Let $G = \mathrm{SU}(N)$ and $|\beta| < 1/(8d)$, or $G = \mathrm{SO}(N)$ with $N \geq 3$ and $|\beta| < (N-2)/(16dN)$. Then the finite-volume Wilson measures satisfy the Bakry–Émery condition with a constant $K > 0$ independent of the volume, the boundary condition and the configuration, and therefore a log-Sobolev and a Poincaré inequality with constants independent of the volume and the boundary condition. Every infinite-volume limit has exponentially decaying correlations of local functions.*

At $d = 4$ the regime is $\beta_W < 1/8$ for SU(2) and $\beta_W < 9/32$ for SU(3) in the usual Wilson normalisation $\beta_W = N^2\beta$, against $\beta_W < 1/12$ and $\beta_W < 3/16$ from (1).

Theorem C (Limit of the curvature method)**.** *For $G = \mathrm{SU}(N)$ on any even torus,*

$$\inf_{Q} \lambda_{\min}\big(\mathrm{Ric} - N\beta\,\mathrm{Hess}\, S_0(Q)\big) = \tfrac{N}{2} - 4dN|\beta|.$$

No Bakry–Émery argument with a lower bound uniform over configurations, in the bi-invariant metric, reaches beyond $|\beta| = 1/(8d)$.

For SU(2) a different route goes further. The conditional law of a single link is a von Mises–Fisher law on the three-sphere, and its spectral gap can be bounded below uniformly in the concentration.

Theorem D (Uniqueness for SU(2))**.** *For $G = \mathrm{SU}(2)$ and $|\beta| < 1/(8(d-1))$ the Wilson theory on $\mathbb{Z}^d$ has exactly one infinite-volume Gibbs measure. At $d = 4$ this is $\beta_W < 1/6$. Single-link Wasserstein–Dobrushin arguments, in the chordal or the geodesic metric, cannot go beyond $|\beta| = 1/(6(d-1))$, that is $\beta_W = 2/9$ at $d = 4$.*

The second purpose is structural. A finite computation at one coupling implies decay at all scales for spin systems by the Dobrushin–Shlosman theorem [12]. For gauge fields the natural boundary data are holonomies, gauge-equivalent boundary conditions must be identified, and the relevant disagreement is the one that remains after an interior gauge transformation. Section 6 sets this up and proves:

Theorem E (Gauge-invariant finite-box criterion)**.** *For every compact group G and every finite-range gauge-invariant interaction, the single finite-box inequality of Definition 6.3 implies uniqueness of the infinite-volume Gibbs measure, exponential decay of all truncated correlations, and the same decay on every sufficiently large torus, uniformly in its size and in every 't Hooft twist sector. The box-hereditary form implies Wilson's area law.*

Two-dimensional SU(2) theory satisfies this criterion at every coupling (Proposition 6.8). For four-dimensional U(1) theory at weak coupling the criterion must fail, because the theory is massless there by the theorems of Guth [16] and Fröhlich–Spencer [15]; this is a consistency requirement that the criterion meets by construction.

Section 7 records the reduction of a gap at fixed coupling to a bound on finite tori that is uniform in their size, together with two structural differences between SU(2) and U(1), and Section 8 states the one problem that separates all of this from a mass gap at intermediate coupling. Nothing in this paper proves a mass gap for four-dimensional Yang–Mills theory, on the lattice at intermediate coupling or in the continuum. The authors' earlier paper [6] shows that no inequality uniform in the lattice spacing, stated in lattice units, can produce the continuum gap; the results here concern fixed spacing.

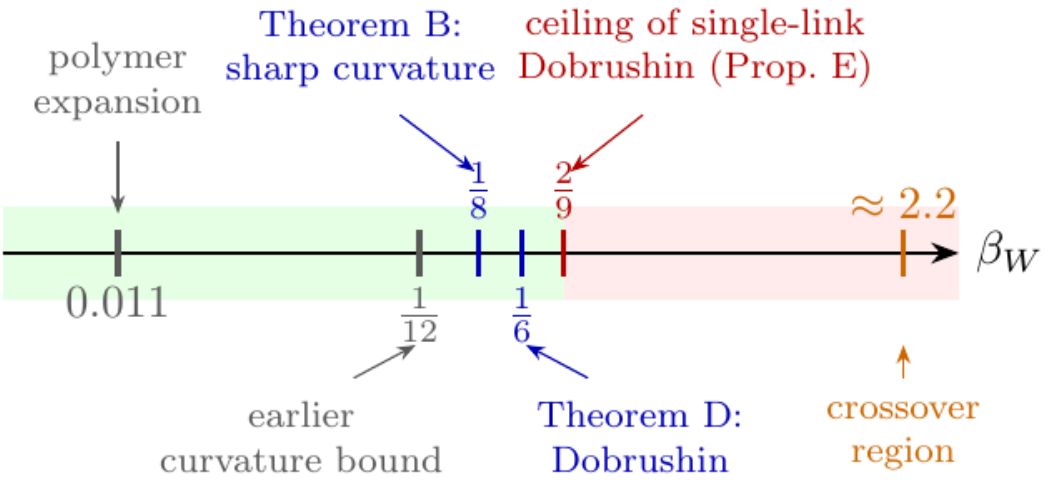


FIGURE 1. Strong-coupling thresholds for four-dimensional SU(2) in the Wilson normalisation β_W (logarithmic scale). Grey: the polymer expansion and the bound (1). Blue: Theorems B and D. Red: the ceiling of single-link Wasserstein–Dobrushin arguments. Orange: the region where lattice simulations place the crossover to weak coupling.[1]

Figure 1 places the thresholds on one axis.

2. SETTING

Let $G \in \{\mathrm{SU}(N), \mathrm{SO}(N), \mathrm{U}(N)\}$ with Lie algebra $\mathfrak{g}$. We use the inner product $\langle X, Y\rangle = -\operatorname{Tr}(XY) = \operatorname{Re}\operatorname{Tr}(X^*Y)$ on $\mathfrak{g}$ and the bi-invariant metric it induces on G. In this metric

$$\mathrm{Ric}_{\mathrm{SU}(N)} = \tfrac{N}{2}, \qquad \mathrm{Ric}_{\mathrm{SO}(N)} = \tfrac{N-2}{4}, \tag{2}$$

since $\mathrm{Ric}(X,X) = -\frac{1}{4}B(X,X)$ for the Killing form B, and $B(X,X) = 2N\operatorname{Tr}X^2$ on $\mathfrak{su}(N)$, $B(X,X) = (N-2)\operatorname{Tr}X^2$ on $\mathfrak{so}(N)$. The group SU(2) with this metric is the round three-sphere of radius $\sqrt{2}$.

Let $\Lambda \subset \mathbb{Z}^d$ be a finite box or a torus, E its set of positively oriented links and P its set of plaquettes. A plaquette p with base point x in the plane of the directions $\mu < \nu$ has the links

$$e_1 = (x,\mu), \quad e_2 = (x+\hat{\mu},\nu), \quad e_3 = (x+\hat{\nu},\mu), \quad e_4 = (x,\nu),$$

and for $Q \in G^E$ we write $Q_p = Q_{e_1}Q_{e_2}Q_{e_3}^*Q_{e_4}^*$. The Wilson measure is

$$d\mu_{\Lambda,\beta}(Q) = Z^{-1}\exp\big(S(Q)\big)\prod_e dQ_e, \qquad S = N\beta\, S_0, \qquad S_0(Q) = \sum_{p\in P}\operatorname{Re}\operatorname{Tr}Q_p, \tag{3}$$

with normalised Haar measure dQ_e; on a box the links outside Λ are frozen to a boundary condition. This is the normalisation of [25]. The Wilson coupling of the physics literature is $\beta_W = N^2\beta$.

A tangent vector at Q is $v = (X_eQ_e)_e$ with $X_e \in \mathfrak{g}$, and $|v|^2 = \sum_e |X_e|^2$. Since $t \mapsto e^{tX}Q$ is a geodesic, $\operatorname{Hess}S_0(v,v) = \frac{d^2}{dt^2}S_0\big((e^{tX_e}Q_e)_e\big)|_{t=0}$.

[1]The value $\beta_W \approx 2.2$ comes from Monte Carlo simulations of the string tension and glueball masses; it plays no role in any proof.

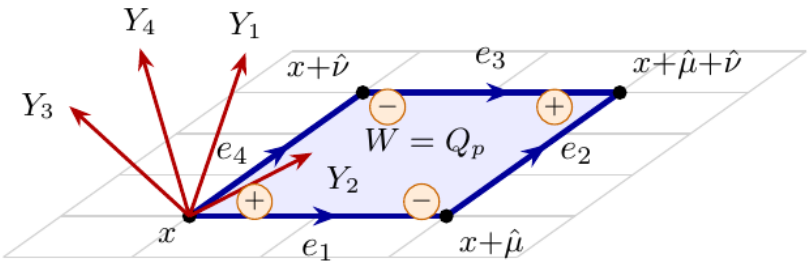


FIGURE 2. A plaquette with its links $e_1, \dots, e_4$, the link vectors $Y_1, \dots, Y_4$ transported to the base point x, and the four corners. The sign at a corner is the sign with which the product of the two link vectors meeting there enters $|D_y X|^2$ at that vertex y, and hence the sign of the corresponding term of the corner form (7).

3. The sharp Hessian bound

3.1. **The Hessian of one plaquette.** Fix a plaquette p and a tangent vector $v = (X_e Q_e)$. Transport the four link vectors to the base point of p:

$$Y_1 = X_{e_1}, \quad Y_2 = Q_{e_1} X_{e_2} Q_{e_1}^*, \quad Y_3 = Q_{e_4} X_{e_3} Q_{e_4}^*, \quad Y_4 = X_{e_4}, \qquad W = Q_p. \tag{4}$$

Then $e^{tX_1} Q_1 e^{tX_2} Q_2 = e^{tY_1} e^{tY_2} Q_1 Q_2$ and $Q_3^* e^{-tX_3} Q_4^* e^{-tX_4} = Q_3^* Q_4^* e^{-tY_3} e^{-tY_4}$, so the perturbed plaquette variable is $Q_p(t) = e^{tY_1} e^{tY_2} W e^{-tY_3} e^{-tY_4}$. Expanding to second order in t and using the cyclicity of the trace,

$$\begin{aligned} f_p &:= \frac{d^2}{dt^2} \operatorname{Re} \operatorname{Tr} Q_p(t)\Big|_{t=0} \\ &= \sum_{i=1}^{4} \operatorname{Re} \operatorname{Tr}(Y_i^2 W) + 2 \operatorname{Re} \operatorname{Tr}(Y_1 Y_2 W) + 2 \operatorname{Re} \operatorname{Tr}(Y_3 Y_4 W) \\ &\quad - 2 \sum_{\substack{i \in \{1,2\} \\ j \in \{3,4\}}} \operatorname{Re} \operatorname{Tr}(Y_i W Y_j), \end{aligned} \tag{5}$$

and $\operatorname{Hess} S_0(v, v) = \sum_p f_p$.

3.2. **The covariant divergence.** The covariant divergence of X at a vertex x is

$$D_x X = \sum_{\mu=1}^{d} \left[X_{x,\mu} - Q_{x-\hat{\mu},\mu}^* X_{x-\hat{\mu},\mu} Q_{x-\hat{\mu},\mu} \right] \in \mathfrak{g}. \tag{6}$$

Each summand is the vector of a link with an end at x, transported to x, with sign $+$ for an outgoing and $-$ for an incoming link. The corner form of the plaquette p is

$$c_p = 2\langle Y_1, Y_4 \rangle - 2\langle Y_1, Y_2 \rangle + 2\langle Y_2, W Y_3 W^* \rangle - 2\langle Y_3, Y_4 \rangle. \tag{7}$$

Lemma 3.1. *For every configuration Q and every finitely supported X,*

$$\sum_x |D_x X|^2 = 2|X|^2 + \sum_p c_p + \sum_{x,\mu} \kappa_{x,\mu}, \qquad \kappa_{x,\mu} = -2\langle X_{x,\mu}, Q_{x-\hat{\mu},\mu}^* X_{x-\hat{\mu},\mu} Q_{x-\hat{\mu},\mu} \rangle,$$

and $\sum_{x,\mu} |\kappa_{x,\mu}| \le 2|X|^2$.

Proof. The vector D_xX is a signed sum of $2d$ vectors, one for each end of a link at x. Expanding $|D_xX|^2$ and summing over x gives three kinds of term. The squares contribute $2|X|^2$, because every link has two ends. The cross term of the outgoing and the incoming end in the same direction μ is $\kappa_{x,\mu}$. The cross terms of two ends in perpendicular directions $\mu \neq \nu$ are in bijection with the pairs consisting of a plaquette in the plane $\mu\nu$ and a corner of it at x, since through x pass four plaquettes of that plane, one for each choice of the two signs.

It remains to identify the four corner terms of a plaquette p with base point x (Figure 2). At x both links are outgoing, which gives $2\langle X_{e_1}, X_{e_4}\rangle = 2\langle Y_1, Y_4\rangle$. At $x+\hat{\mu}$ the end of e_1 is incoming and e_2 is outgoing, which gives $-2\langle Q_{e_1}^* X_{e_1} Q_{e_1}, X_{e_2}\rangle = -2\langle Y_1, Y_2\rangle$ after conjugation by Q_{e_1}. In the same way the corner $x+\hat{\nu}$ gives $-2\langle Y_4, Y_3\rangle$. At $x+\hat{\mu}+\hat{\nu}$ both links are incoming, which gives $2\langle Q_{e_2}^* X_{e_2} Q_{e_2}, Q_{e_3}^* X_{e_3} Q_{e_3}\rangle$; conjugating by $Q_{e_1}Q_{e_2}$ and using $Q_{e_1}Q_{e_2}Q_{e_3}^* = WQ_{e_4}$ turns it into $2\langle Y_2, WY_3W^*\rangle$. The four terms add up to c_p. Finally, $|\kappa_{x,\mu}| \leq |X_{x,\mu}|^2 + |X_{x-\hat{\mu},\mu}|^2$ by the Cauchy–Schwarz inequality, and summing over (x,μ) counts every link twice. □

3.3. **A local inequality.**

Lemma 3.2. *For every N, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$,*

$$f_p + c_p \leq 2\sum_{i=1}^{4} |Y_i|^2 \qquad \text{and} \qquad -f_p + c_p \leq 2\sum_{i=1}^{4} |Y_i|^2.$$

Proof. Replacing W by $-W$ changes the sign of f_p and leaves c_p unchanged, so the second inequality follows from the first. Write $W = VDV^*$ with V unitary and $D = \mathrm{diag}(w_1, \dots, w_N)$, $|w_a| = 1$. Replacing every Y_i by V^*Y_iV leaves both sides unchanged, so we may assume $W = D$. Every term of f_p and c_p then has one of the forms

$$\mathrm{Re}\sum_{a,b}(Y_i)_{ab}(Y_j)_{ba}w_a, \quad \mathrm{Re}\sum_{a,b}(Y_i)_{ab}w_b(Y_j)_{ba}, \quad \mathrm{Re}\sum_{a,b}\overline{(Y_i)_{ab}}(Y_j)_{ab}, \quad \mathrm{Re}\sum_{a,b}\overline{(Y_i)_{ab}}(Y_j)_{ab}w_a\bar{w}_b.$$

Since $(Y_i)_{ba} = -\overline{(Y_i)_{ab}}$, each of them couples only the entries in the positions (a,b) and (b,a). Hence the quadratic form $F = 2\sum|Y_i|^2 - f_p - c_p$ is an orthogonal sum of blocks of two kinds: one for each diagonal position a, in the variables $(Y_i)_{aa} = \mathrm{i}y_i$ with $y \in \mathbb{R}^4$, and one for each unordered pair $a \neq b$, in the variables $z_i = (Y_i)_{ab} \in \mathbb{C}$. It suffices to show that each block is nonnegative.

Diagonal blocks. Put $w_a = e^{\mathrm{i}\varphi}$. Substituting $Y_i = \mathrm{i}y_i$, $W = e^{\mathrm{i}\varphi}$ in (5) and (7) and expanding gives

$$F_{aa} = (y_1+y_3)^2 + (y_2+y_4)^2 + (1+\cos\varphi)(y_1+y_2-y_3-y_4)^2 \geq 0.$$

Pair blocks. Put $w_a = e^{\mathrm{i}(\chi+\theta)}$, $w_b = e^{\mathrm{i}(\chi-\theta)}$ and $m = \cos\chi$. Since $|Y_i|^2$ contains $2|z_i|^2$ from the two positions $(a,b),(b,a)$, the block is $F_{ab} = z^*Mz$ with $z = (z_1,\dots,z_4)$ and the Hermitian matrix M whose diagonal entries are $4 + 2m\cos\theta$ and whose entries above the diagonal are

$$M_{12} = M_{34} = 2 + 2me^{-\mathrm{i}\theta}, \quad M_{13} = M_{24} = -2me^{\mathrm{i}\theta},$$
$$M_{14} = -2 - 2me^{\mathrm{i}\theta}, \quad M_{23} = -2e^{2\mathrm{i}\theta} - 2me^{\mathrm{i}\theta}.$$

These are obtained by expanding the eight terms of (5) and (7) on the 2×2 minor in the rows and columns a, b. Every entry is affine in m, and a direct comparison shows $M(-1,\theta) = M(1,\theta+\pi)$. Hence $M(m,\theta) = \frac{1+m}{2}M(1,\theta) + \frac{1-m}{2}M(1,\theta+\pi)$, and it suffices to show $M(1,\theta) \succeq 0$ for all θ. Substitute $z = \Delta u$ with $\Delta = \mathrm{diag}(1, e^{\mathrm{i}\theta/2}, e^{-\mathrm{i}\theta}, e^{-\mathrm{i}\theta/2})$ and put $c = \cos(\theta/2)$. The matrix

$\Delta^* M(1,\theta)\Delta$ is real:

$$\Delta^* M(1,\theta)\Delta = \begin{pmatrix} 2+4c^2 & 4c & -2 & -4c \\ 4c & 2+4c^2 & -4c & -2 \\ -2 & -4c & 2+4c^2 & 4c \\ -4c & -2 & 4c & 2+4c^2 \end{pmatrix},$$

and comparing coefficients shows that the corresponding form is

$$2c^2(|u_1+u_3|^2+|u_2+u_4|^2)+(1+c)^2|u_1+u_2-u_3-u_4|^2+(1-c)^2|u_1-u_2-u_3+u_4|^2 \;\geq\; 0.$$

For instance, the diagonal coefficient is $2c^2+(1+c)^2+(1-c)^2 = 2+4c^2$ and the $(1,3)$ coefficient is $2c^2-(1+c)^2-(1-c)^2 = -2$. □

Remark 3.3. The blocks of the ordered positions (a,b) and (b,a) taken separately are not positive semidefinite (take $w_a=-1$, $w_b=1$). The pairing of (a,b) with (b,a) through $(Y)_{ba} = -\overline{(Y)_{ab}}$ is what makes the inequality true.

3.4. **Proof of Theorem A.**

Proof of the upper bound. Every link lies in $2(d-1)$ plaquettes and $|Y_i| = |X_{e_i}|$, so summing Lemma 3.2 over p gives $\sum_p(\pm f_p + c_p) \leq 4(d-1)|X|^2$. By Lemma 3.1, $\sum_p c_p = \sum_x |D_x X|^2 - 2|X|^2 - \sum \kappa \geq \sum_x |D_x X|^2 - 4|X|^2$. Hence

$$\pm \operatorname{Hess} S_0(v,v) = \pm \sum_p f_p \leq 4(d-1)|X|^2 + 4|X|^2 - \sum_x |D_x X|^2. \qquad \square$$

Proof of sharpness. Let the torus have even side length, and let $X_{x,\mu} = (-1)^{|x|}\varepsilon_\mu T$ with $|x| = \sum_\nu x_\nu$, a vector $\varepsilon \in \mathbb{R}^d$ with $\sum_\mu \varepsilon_\mu = 0$, and $T \in \mathfrak{g}$.

The sign −. At the identity configuration $W=1$ and $Y_i = X_{e_i}$, and (5) becomes $f_p = \operatorname{Tr}((Y_1+Y_2-Y_3-Y_4)^2) = -|Y_1+Y_2-Y_3-Y_4|^2$. For the plaquette at x in the plane $\mu\nu$, $Y_1+Y_2-Y_3-Y_4 = 2(-1)^{|x|}(\varepsilon_\mu - \varepsilon_\nu)T$. Each vertex therefore contributes $-4|T|^2\sum_{\mu<\nu}(\varepsilon_\mu-\varepsilon_\nu)^2 = -4|T|^2\big(d|\varepsilon|^2 - (\sum_\mu \varepsilon_\mu)^2\big) = -4d|\varepsilon|^2|T|^2$, while it carries $|\varepsilon|^2|T|^2$ of $|X|^2$. So $\operatorname{Hess} S_0(v,v) = -4d|X|^2$. Moreover $D_x X = 2(-1)^{|x|}(\sum_\mu \varepsilon_\mu)T = 0$, so the inequality of Theorem A is an equality.

The sign +. Let $z = \operatorname{diag}(-1,-1,1,\dots,1)$, which lies in $\mathrm{SU}(N)$, $\mathrm{SO}(N)$ and $\mathrm{U}(N)$, and let $Q_{x,\mu} = z^{\sum_{\lambda<\mu} x_\lambda}$. This is well defined on an even torus because $z^2 = 1$. For the plaquette at x in the plane $\mu<\nu$ one finds $Q_{e_1} = Q_{e_3} = z^{\sum_{\lambda<\mu} x_\lambda}$, $Q_{e_4} = z^{\sum_{\lambda<\nu} x_\lambda}$ and $Q_{e_2} = zQ_{e_4}$, so every plaquette variable equals z. Take the same X with $T \neq 0$ supported in the upper left 2×2 block, so that T commutes with z and $zT = -T$. All transports in (4) and (6) act trivially on T, every term of (5) changes sign relative to the identity configuration, and therefore $\operatorname{Hess} S_0(v,v) = +4d|X|^2$ and again $D_x X = 0$. □

4. Consequences of the curvature bound

Proof of Theorem B. By Theorem A, $|\operatorname{Hess} S(v,v)| = N|\beta||\operatorname{Hess} S_0(v,v)| \leq 4dN|\beta||v|^2$. The Ricci curvature of the product manifold G^E, or of $G^{E_{\mathrm{int}}}$ with the boundary links frozen, equals Ric_G in every direction. By (2),

$$\mathrm{Ric} - \operatorname{Hess} S \;\geq\; K := \tfrac{N}{2} - 4dN|\beta| \quad (\mathrm{SU}(N)), \qquad K := \tfrac{N-2}{4} - 4dN|\beta| \quad (\mathrm{SO}(N)),$$

uniformly in the volume, the boundary condition and the configuration, and $K>0$ in the stated ranges. The Bakry–Émery criterion on a compact Riemannian manifold [2, Cor. 5.7.2] gives the log-Sobolev inequality $\mathrm{Ent}_\mu(f^2) \leq \frac{2}{K}\mu|\nabla f|^2$ and the Poincaré inequality with constant $1/K$.

For the decay of correlations we follow [25, §4]. The argument there uses the Langevin dynamics of the Wilson measure and the gradient bound $|\nabla P_t f| \le e^{-Kt} P_t |\nabla f|$, which is equivalent to the Bakry–Émery condition with constant K; the Hessian bound enters only through the value of that constant. With K in place of the constant $K_S = \frac{N}{2} - 8N|\beta|(d-1)$ of [25], the proofs of [25, Cor. 4.11 and Rem. 4.12] apply without change and give exponential decay of the covariance of local functions, uniformly in the volume, for every infinite-volume limit. □

Remark 4.1. For SU(2), uniqueness of the infinite-volume measure in the whole range of Theorem B follows from Theorem D, whose range $1/(8(d-1))$ is larger. For SU(N) with $N \ge 3$ and for SO(N) the uniqueness argument of [25, §5] uses a separate estimate with its own constant, and we do not claim uniqueness on the whole range $|\beta| < 1/(8d)$.

Proof of Theorem C. By Theorem A, $(\mathrm{Ric} - N\beta \operatorname{Hess} S_0)(v,v) \ge (\frac{N}{2} - 4dN|\beta|)|v|^2$ for every configuration and every v, so the infimum is at least $\frac{N}{2} - 4dN|\beta|$. For the reverse inequality we may assume $\beta \ne 0$ and take the vector field v of the proof of sharpness. If $\beta > 0$ we use the frustrated configuration, at which $\operatorname{Hess} S_0(v,v) = 4d|v|^2$; if $\beta < 0$ we use the identity configuration, at which $\operatorname{Hess} S_0(v,v) = -4d|v|^2$. In both cases $N\beta \operatorname{Hess} S_0(v,v) = 4dN|\beta||v|^2$. The bi-invariant metric on SU(N) is Einstein, so $\mathrm{Ric}(v,v) = \frac{N}{2}|v|^2$, and therefore $(\mathrm{Ric} - N\beta \operatorname{Hess} S_0)(v,v) = (\frac{N}{2} - 4dN|\beta|)|v|^2$. □

A constant rescaling of the metric on all links multiplies Ric and Hess S by the same factor, and a link-dependent constant rescaling does not change the infimum on a torus, by translation invariance of the extremal configurations. Theorem C is a statement about the method; it does not say that anything changes in the theory at $|\beta| = 1/(8d)$.

5. Uniqueness for SU(2)

5.1. **The single-link law.** Identify SU(2) with the unit quaternions, so that $\operatorname{Re}\operatorname{Tr} U = 2x_0(U)$. Fix a link e and all other links. The action depends on Q_e through $N\beta \operatorname{Re}\operatorname{Tr}(Q_e A)$, where A is the sum of the $2(d-1)$ staples of e, each an element of SU(2), so A corresponds to a vector of $\mathbb{R}^4$ of length at most $2(d-1)$. The conditional law of Q_e is therefore a von Mises–Fisher law

$$\nu_\kappa(dx) \propto e^{\kappa x_1} \sigma(dx) \qquad \text{on the unit sphere } S^3 \subset \mathbb{R}^4,$$

in coordinates in which the staple direction is the x_1 axis, with concentration $\kappa = 4|\beta||A|$ (and x_1 replaced by $-x_1$ if $\beta < 0$).

Lemma 5.1. *For every $\kappa \ge 0$ the spectral gap of ν_κ on the unit sphere S^3 is at least 3, with equality at $\kappa = 0$. On the sphere of radius $\sqrt{2}$, which is* SU(2) *in the metric of Section 2, it is at least 3/2.*

Proof. Let $\theta \in (0,\pi)$ be the polar angle from the x_1 axis. The generator of the reversible diffusion for ν_κ is

$$L = \partial_\theta^2 + b\,\partial_\theta + \sin^{-2}\theta\,\Delta_{S^2}, \qquad b = 2\cot\theta - \kappa\sin\theta = \frac{w'}{w}, \quad w = \sin^2\theta\, e^{\kappa\cos\theta}.$$

On spherical harmonics of degree ℓ on S^2 it acts as the Sturm–Liouville operator $L_\ell = w^{-1}\partial_\theta(w\partial_\theta) - \ell(\ell+1)\sin^{-2}\theta$ on $L^2((0,\pi), w\,d\theta)$.

We use the ground-state bound of Barta [5]. Let $\mathcal{L} = w^{-1}\partial(w\partial) - q$ with q continuous on $(0,\pi)$, let $\psi > 0$ be smooth on $(0,\pi)$ with $-\mathcal{L}\psi \ge c\psi$, and let f be smooth with compact

support in $(0,\pi)$. Writing $f=\psi h$ and integrating by parts,

$$\int_0^\pi (f'^2+qf^2)w\,d\theta=\int_0^\pi \psi^2 h'^2 w\,d\theta+\int_0^\pi(-\mathcal{L}\psi)\psi h^2 w\,d\theta \;\geq\; c\int_0^\pi f^2 w\,d\theta.$$

When $q\geq 2\sin^{-2}\theta - C$ near the endpoints, compactly supported functions form a core for the quadratic form, and the bound holds for the lowest point of the spectrum of $-\mathcal{L}$.

Degrees $\ell\geq 1$. Here $q=\ell(\ell+1)\sin^{-2}\theta$ increases with ℓ, so it suffices to take $\ell=1$. With $\psi=\sin\theta\, e^{a\kappa\cos\theta}$ a direct differentiation gives

$$-\frac{L_1\psi}{\psi}=3+(5a+1)\kappa\cos\theta-a(a+1)\kappa^2\sin^2\theta,$$

and for $a=-\frac{1}{5}$ this equals $3+\frac{4}{25}\kappa^2\sin^2\theta\geq 3$.

Degree $\ell=0$. Let g be a nonconstant zonal eigenfunction, $L_0 g=-\lambda g$, and put $h=g'$. Then $h(0)=h(\pi)=0$, and differentiating the eigenvalue equation gives $h''+bh'+b'h=-\lambda h$, that is $\mathcal{M}h=-\lambda h$ with $\mathcal{M}=w^{-1}\partial(w\partial)+b'$ and $-b'=2\sin^{-2}\theta+\kappa\cos\theta$. With the same ψ,

$$-\frac{\mathcal{M}\psi}{\psi}=3+(5a+2)\kappa\cos\theta-a(a+1)\kappa^2\sin^2\theta,$$

and for $a=-\frac{2}{5}$ this equals $3+\frac{6}{25}\kappa^2\sin^2\theta\geq 3$. Hence $\lambda\geq 3$.

At $\kappa=0$ the gap of S^3 is 3. Rescaling the radius from 1 to $\sqrt{2}$ divides eigenvalues by 2. □

5.2. **Proof of Theorem D.** Let ρ be the geodesic distance on SU(2) in the metric of Section 2. For links $e\neq e'$ the Dobrushin coefficient is

$$C_{ee'}=\sup\frac{W_1(\mu_e^{\omega},\mu_e^{\omega'})}{\rho(\omega_{e'},\omega'_{e'})},$$

where μ_e^{ω} is the conditional law of Q_e given the configuration ω off e, W_1 is the Kantorovich distance for ρ, and the supremum runs over pairs ω,ω' that differ only at e'.

Proof. If e and e' do not lie in a common plaquette, μ_e^{ω} does not depend on $\omega_{e'}$ and $C_{ee'}=0$. Otherwise they lie in exactly one common plaquette. Join $\omega_{e'}$ to $\omega'_{e'}$ by a unit-speed geodesic $s\mapsto\omega_{e'}(s)$ and let μ_s be the corresponding conditional law at e. For a 1-Lipschitz function g,

$$\frac{d}{ds}\mu_s(g)=\operatorname{Cov}_{\mu_s}(g,\partial_s S),\qquad |\operatorname{Cov}_{\mu_s}(g,\partial_s S)|\leq(\operatorname{Var} g\,\operatorname{Var}\partial_s S)^{1/2}\leq\tfrac{2}{3}\,\|\nabla_e g\|_{L^2}\,\|\nabla_e\partial_s S\|_{L^2},$$

by the Poincaré inequality of Lemma 5.1. Only the common plaquette contributes to $\nabla_e\partial_s S$; its mixed second derivative is $N\beta\operatorname{Re}\operatorname{Tr}(AWB)$ with unit vectors $A,B\in\mathfrak{su}(2)$ and $W\in\mathrm{SU}(2)$, so $|\nabla_e\partial_s S|\leq N|\beta|=2|\beta|$. Integrating in s and taking the supremum over g (Kantorovich–Rubinstein duality) gives $C_{ee'}\leq\frac{4}{3}|\beta|$. A link lies in $2(d-1)$ plaquettes, each with three other links, so

$$\alpha:=\sup_e\sum_{e'}C_{ee'}\leq 6(d-1)\cdot\tfrac{4}{3}|\beta|=8(d-1)|\beta|.$$

For $\alpha<1$ Dobrushin's uniqueness theorem for compact metric spin spaces with the Kantorovich distance [11, 13] gives uniqueness of the infinite-volume Gibbs measure. □

Remark 5.2. Dobrushin's condition also gives decay of correlations of Lipschitz local functions at rate $\log(1/\alpha)$ per unit distance [14]. For SU(N) with $N\geq 3$ the staple sum is not a multiple of a unitary matrix, and the same argument with the single-link Bakry–Émery bound reproduces only $|\beta|<1/(16(d-1))$.

Proposition 5.3 (The ceiling)**.** *Let δ be the chordal metric of $S^3 \subset \mathbb{R}^4$ or the geodesic metric, used for both the distance at e' and the Kantorovich distance at e. For all links e, e' in a common plaquette, $C_{ee'} \geq |\beta|$. Hence no argument through single-link Dobrushin coefficients proves uniqueness beyond $|\beta| = 1/(6(d-1))$.*

Proof. The $2(d-1)$ staples of e involve pairwise disjoint sets of links other than e', so they can be chosen with sum zero while the staple through e' is an arbitrary unit quaternion. Then $\kappa = 0$ and μ_e^ω is uniform. Moving $Q_{e'}$ by a small distance ε moves the staple sum by an isometric image of the displacement, since left and right multiplications by unit quaternions are isometries of $\mathbb{R}^4$. For $h \in \mathbb{R}^4$ the derivative of the law in the direction h at zero staple is the signed measure $4\beta\langle x, h\rangle\sigma(dx)$. Testing it against the 1-Lipschitz function $x \mapsto \langle x, \hat{h}\rangle$ gives $4|\beta||h| \int \langle x, \hat{h}\rangle^2 d\sigma = |\beta||h|$, since $\int x_1^2 \, d\sigma = \frac{1}{4}$ on S^3. Hence $C_{ee'} \geq |\beta|$, and summing over the $6(d-1)$ neighbours gives $\alpha \geq 6(d-1)|\beta|$. □

Remark 5.4. In the chordal metric the reflection coupling $x \mapsto x - 2\langle x, \hat{h}\rangle\hat{h}$ shows that $|\beta|$ is the exact value at zero staple. A 2026 preprint attributed to J. Lockwood, which we have seen only as an abstract, announces uniqueness for four-dimensional SU(2) at $\beta_W < 2/9$ by the Wasserstein–Dobrushin method. By Proposition 5.3 that value is exactly the ceiling of the method, and it would follow if the Lipschitz constant of the von Mises–Fisher family on S^3 were largest at zero concentration. We have not been able to check that claim.

6. A gauge-invariant finite-box criterion

6.1. **Interactions and boundary data.** Let G be a compact metrisable group with normalised Haar measure λ, and let E be the set of positively oriented links of $\mathbb{Z}^d$ or of a torus. An interaction is a family $\Phi = (\Phi_X)$ indexed by finite sets $X \subset E$ of diameter at most R, with $\Phi_X \colon G^X \to \mathbb{R}$ continuous, invariant under translations, and gauge invariant: $\Phi_X(gU) = \Phi_X(U)$ for every $g \colon \mathbb{Z}^d \to G$, where $(gU)_{xy} = g_x U_{xy} g_y^{-1}$. The Wilson action is $\Phi_p = -N\beta \operatorname{Re}\operatorname{Tr} U_p$; a renormalised effective action adds further terms.

For finite $V \subset E$ let $\partial V = \bigcup\{X \setminus V : X \cap V \neq \emptyset,\ \Phi_X \not\equiv 0\}$. The specification is

$$\gamma_V(dU_V \mid \omega) = Z_V(\omega)^{-1} \exp\Big(-\sum_{X \cap V \neq \emptyset} \Phi_X(U_V \omega)\Big) \lambda^V(dU_V),$$

and it depends on ω only through $\omega_{\partial V}$. Let $I(V)$ be the set of vertices all of whose incident links lie in V, and $\mathcal{G}_V = G^{I(V)}$. A gauge transformation at $x \in I(V)$ moves only links of V, so

$$g_* \gamma_V(\cdot \mid \omega) = \gamma_V(\cdot \mid \omega) \qquad (g \in \mathcal{G}_V). \tag{8}$$

Lemma 6.1. *Let Γ_V be the graph with edge set ∂V. If $\omega'_{\partial V} = (h\omega)_{\partial V}$ for a map h from the vertices of Γ_V to G, then $\gamma_V(\cdot \mid \omega') = (\tilde{h}|_V)_* \gamma_V(\cdot \mid \omega)$ for any extension $\tilde{h}$ of h. Consequently gauge-invariant expectations $\gamma_V(F \mid \omega)$ depend only on the class of $\omega_{\partial V}$ in $G^{\partial V}/G^{\mathrm{vert}(\Gamma_V)}$, which for each connected component of Γ_V is the space of holonomy representations of its fundamental group modulo conjugation. (This is the holonomy of the gauge field; for the holonomy of a Riemannian metric in a related $\mathrm{SU}(n)$ setting see* [8]*.)*

Proof. Change variables $U_V \mapsto \tilde{h} U_V$: Haar measure and the energy are invariant. For the second statement, gauge the links of a spanning tree of each component to the identity; the remaining links are free generators of the fundamental group, and the residual freedom is one constant per component, acting by conjugation. □

Plaquette holonomies alone do not determine this class when G is non-abelian; holonomies of larger loops of the shell enter.[2] The criterion below therefore compares boundary conditions that differ at one link.

Lemma 6.2 (Gauge lift)**.** *Let Π be a coupling of $\gamma_V(\cdot \mid \omega)$ and $\gamma_V(\cdot \mid \omega')$, let $g\colon G^V \times G^V \to \mathcal{G}_V$ be measurable, and let h be Haar distributed on $\mathcal{G}_V$, independent of $(U, U') \sim \Pi$. Then $(hU,\, hg(U,U')U')$ is again a coupling of the two measures, and $\mathbb{P}[(hU)_b \neq (hgU')_b] = \Pi[U_b \neq (g(U,U')U')_b]$ for every link b.*

Proof. By (8), $hU \sim \gamma_V(\cdot \mid \omega)$. Given (U, U'), the element hg is Haar distributed, so hgU' is uniform on the $\mathcal{G}_V$-orbit of U', and by (8) again $hgU' \sim \gamma_V(\cdot \mid \omega')$. Each h_x acts on a link by a bijection of G, which preserves equality of the two link values. □

6.2. **The criterion.** Write V_μ for the links of V in direction μ, $n_{\min} = \min_\mu |V_\mu|$, and $\omega \sim_y \omega'$ when ω, ω' differ only at the link $y \in \partial V$.

Definition 6.3. The interaction satisfies the gauge Dobrushin–Shlosman condition $\mathrm{GDS}(V, \varepsilon)$ if there are a Borel family of couplings $\Pi_{\omega,\omega'}$ of $\gamma_V(\cdot \mid \omega)$ and $\gamma_V(\cdot \mid \omega')$, for all pairs $\omega \sim_y \omega'$, and numbers

$$\kappa_V(y, b) \;\geq\; \sup_{\omega \sim_y \omega'} \Pi_{\omega,\omega'}[U_b \neq U'_b] \qquad (y \in \partial V,\ b \in V)$$

such that for every direction ν

$$K_V(\nu) := \sum_{y \in \partial V,\ \dim y = \nu} \sum_{b \in V} \kappa_V(y, b) \;\leq\; (1 - \varepsilon)\, n_{\min}. \tag{9}$$

By Lemmas 6.1 and 6.2 the couplings may be constructed on the gauge quotient and then aligned by any measurable choice of interior gauge, and the supremum is a supremum over classes of shell holonomies (Figure 3). For a box symmetric under the hypercubic group, (9) reads $\sum_{y,b} \kappa_V(y, b) \leq (1 - \varepsilon)|V|$, the form of Dobrushin and Shlosman [12], except that the supremum over boundary conditions is taken for each b separately; this is what makes the proof below work for an arbitrary compact group.

Theorem 6.4. *Assume $\mathrm{GDS}(V, \varepsilon)$ for a finite-range gauge-invariant interaction with compact G. Let $D = \max\{|b - y| : b \in V,\ y \in \partial V\}$, with distances between link midpoints, and $0 < \lambda < |\log(1 - \varepsilon)|/D$, so that $\theta := (1 - \varepsilon)e^{\lambda D} < 1$. Then:*

- (i) *There is exactly one Gibbs measure μ on G^E; it is invariant under translations and gauge transformations.*
- (ii) *Let $S \subset E$ be finite and let μ_1, μ_2 be probability measures invariant under γ_{V+a} for every translate with $(V + a) \cap S = \emptyset$. Let N_S be S together with the links of the translates of V that meet S; then $|N_S| \leq c_V|S|$. There is a coupling of μ_1 and μ_2 under which, for every link b at distance greater than $\operatorname{diam} V + R$ from S,*

$$\mathbb{P}[\sigma_b \neq \sigma'_b] \leq \frac{\theta}{1 - \theta} \sum_{z \in N_S} e^{-\lambda|b - z|}.$$

- (iii) *For bounded F, H with finite supports S_F, S_H,*

$$|\operatorname{Cov}_\mu(F, H)| \leq 2\|F\|_\infty \|H\|_\infty \frac{\theta}{1 - \theta}\, c_V |S_F||S_H|\, e^{-\lambda(\operatorname{dist}(S_F, S_H) - \operatorname{diam} V - R)}.$$

[2] Take two plaquettes of the shell sharing a vertex, with based holonomies (X, Y) and (X, Y'), where $X = Y = \mathrm{i}\sigma_3$ and $Y' = \mathrm{i}\sigma_1$ in $\mathrm{SU}(2)$. Each holonomy is conjugate to its partner, but $\operatorname{Tr}(XY) = -2 \neq 0 = \operatorname{Tr}(XY')$, and XY is the holonomy of the boundary of the union of the two plaquettes.

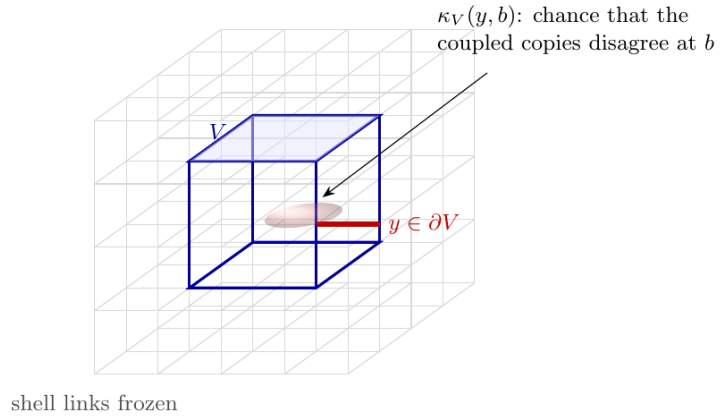


FIGURE 3. The box V and a boundary link y. Two copies of the box are coupled, with boundary conditions differing only at y; $\kappa_V(y,b)$ bounds the probability that they disagree at b after the best interior gauge alignment.

(iv) *Items* (i)–(iii) *hold on every torus of side greater than* $2(\operatorname{diam} V + R)$ *with the same constants, for the untwisted and for every 't Hooft twisted Wilson action.*

Proof. A minimal coupling. Let W be E or $E \setminus S$, let $\mathcal{A} = \{a : V + a \subseteq W\}$ and $V_a = V + a$. Fix a link b_0 and put $w_b = e^{-\lambda|b-b_0|}$, so that $\sum_b w_b < \infty$. On the set $\mathcal{C}$ of couplings of μ_1, μ_2 define $\Psi(\pi) = \sum_b w_b\, \pi[\sigma_b \neq \sigma'_b]$. The set $\mathcal{C}$ is weakly compact because G^E is compact, and each map $\pi \mapsto \pi[\sigma_b \neq \sigma'_b]$ is lower semicontinuous, being the measure of an open set. So Ψ attains its minimum at some π^*. Put $\rho(b) = \pi^*[\sigma_b \neq \sigma'_b]$.

Block resampling. For $a \in \mathcal{A}$ define a new coupling $T_a\pi^*$: sample $(\sigma, \sigma') \sim \pi^*$, keep both configurations off V_a, and inside V_a sample from the following path coupling of $\gamma_{V_a}(\cdot \mid \sigma)$ and $\gamma_{V_a}(\cdot \mid \sigma')$. List $\partial V_a = \{z_1, \dots, z_k\}$ and let ω^j agree with σ' on $z_1, \dots, z_j$ and with σ elsewhere. Consecutive ω^{j-1}, ω^j are equal or differ only at z_j; couple their block laws by the identity in the first case and by the translate of $\Pi_{\omega^{j-1},\omega^j}$ in the second, and chain the couplings through their disintegrations, which depend measurably on the parameters [18, Ch. 3]. This produces $\tau^0, \dots, \tau^k$ with $\tau^0 \sim \gamma_{V_a}(\cdot \mid \sigma)$, $\tau^k \sim \gamma_{V_a}(\cdot \mid \sigma')$ and $\mathbf{1}[\tau^0_b \neq \tau^k_b] \leq \sum_j \mathbf{1}[\tau^{j-1}_b \neq \tau^j_b]$. Since each μ_i is γ_{V_a}-invariant, $T_a\pi^* \in \mathcal{C}$, and for $b \in V_a$

$$T_a\pi^*[\tau_b \neq \tau'_b] \leq \sum_{z \in \partial V_a} \kappa_V(z-a, b-a)\rho(z),$$

while nothing changes off V_a. Minimality, $\Psi(T_a\pi^*) \geq \Psi(\pi^*)$, gives

$$\sum_{b \in V_a} w_b \rho(b) \leq \sum_{b \in V_a} w_b \sum_{z \in \partial V_a} \kappa_V(z-a, b-a)\rho(z). \tag{10}$$

Summation over blocks. Sum (10) over $a \in \mathcal{A}$; all series converge, the left side being at most $|V| \sum_b w_b$. Let W° be the set of links b such that every translate of V containing b lies in W; then $E \setminus W^\circ \subseteq N_S$. A link b of direction μ lies in exactly $|V_\mu| \geq n_{\min}$ translates. On the right, $|z - b| \leq D$ gives $w_b \leq e^{\lambda D} w_z$, and for fixed z of direction ν the sum over all a with $z \in \partial V_a$ and all $b \in V_a$ of $\kappa_V(z-a, b-a)$ equals $K_V(\nu)$. By (9),

$$n_{\min} \sum_{b \in W^\circ} w_b \rho(b) \leq \theta\, n_{\min} \Big[\sum_{z \in W^\circ} w_z \rho(z) + \sum_{z \in N_S} w_z \Big],$$

so $\sum_{W^\circ} w\rho \le \frac{\theta}{1-\theta}\sum_{z\in N_S} w_z$. Taking $b_0 \in W^\circ$ and keeping only the term $b = b_0$ on the left proves (ii).

Uniqueness. With $S = \emptyset$, $N_S = \emptyset$ and $\rho \equiv 0$, so any two Gibbs measures coincide. Gibbs measures exist by compactness, and translates and gauge transforms of a Gibbs measure are Gibbs measures; since there is only one, it is invariant. This is (i).

Decay. Let $\mu^u = \mu(\cdot \mid U_{S_H} = u)$ be a regular conditional probability; for μ-almost every u it is invariant under every γ_{V_a} with $V_a \cap S_H = \emptyset$. Since $\mathrm{Cov}(F,H) = \int H(u)[\mu^u(F) - \int \mu^{u'}(F)\mu(du')]\mu(du)$, $|\mathrm{Cov}(F,H)| \le \|H\|_\infty \operatorname{ess\,sup}_{u,u'} |\mu^u(F) - \mu^{u'}(F)|$, and $|\mu^u(F) - \mu^{u'}(F)| \le 2\|F\|_\infty \sum_{b\in S_F} \rho(b)$. Item (ii) with $S = S_H$ gives (iii).

Tori and twists. On a torus of side greater than $2(\operatorname{diam} V + R)$ every translate is a genuine block and the argument is unchanged. Let $z_p \in Z(G)$ be a twist that is closed, so that on every ball of radius at least $\operatorname{diam} V + R$ it equals $(dn)_p = \prod_{e\in\partial p} n_e^{\pm 1}$ for some $n\colon E \to Z(G)$; every 't Hooft twist has this property, because a closed $Z(G)$-valued 2-cocycle is exact on a contractible ball. Multiplication M_n of each link by the central element n_e satisfies $\gamma^{\mathrm{tw}}_{V_a}(\cdot \mid \omega) = (M_n)_*\gamma_{V_a}(\cdot \mid M_n\omega)$ on such a ball, since the central factors commute through, and M_n is a bijection on each link. So the twisted specification satisfies $\mathrm{GDS}(V,\varepsilon)$ with the same κ_V, and the proof applies. This is (iv). □

6.3. **Confinement and stability.** Let $\mathcal{V}$ be the family of sets $V \cap W'$, where W' runs over the interior link sets of all cubes in all positions. The box-hereditary condition $\mathrm{HGDS}(V,\varepsilon)$ is (9) with $\sum_b \kappa_V(y,b)$ replaced by $\sup_{V'\in\mathcal{V},\ y\in\partial V'} \sum_{b\in V'} \kappa_{V'}(y,b)$, the boundary condition of V' including the links of $V \setminus V'$.

Proposition 6.5. *Assume* $\mathrm{HGDS}(V,\varepsilon)$ *for the Wilson action of a compact group* $G \subset \mathrm{U}(n)$. *Then covariances of local functions under the finite-volume measure of every cube with every boundary condition decay exponentially with constants independent of the cube and the boundary condition. By Chatterjee's theorem* [10, Thm. 2.4]*, the centre symmetry is unbroken and, if the representation acts non-trivially on the centre, Wilson loops obey the area law.*

Proof. Fix a cube B with boundary condition δ and interior link set W_B, and local functions F, H. Condition $\mu_{B,\delta}$ on $U_{S_H} = u$ and on $U_{S_H} = u'$, and run the proof of Theorem 6.4 in $W = W_B \setminus S_H$ with all blocks $V_a \cap W_B$ that meet W. A link of W of direction μ now lies in $|V_\mu| \ge n_{\min}$ blocks, so no boundary layer is lost on the left side of the summed inequality. A link $z \in W$ that lies in $\partial(V_a \cap W_B)$ lies in ∂V_a, and its total weight on the right is bounded by the hereditary condition. The remaining links of $\partial(V_a \cap W_B)$ either lie outside B, where both copies carry δ, or lie in S_H. So the only sources of disagreement are the links of S_H, each with weight at most $(|V| + |\partial V|)|V|/n_{\min}$ times $e^{\lambda D}$, and the covariance bound follows as in Theorem 6.4(iii), uniformly in B and δ. This is the decay property of [10, Def. 2.3]. □

Lemma 6.6. *Suppose* Φ *satisfies* $\mathrm{GDS}(V,\varepsilon)$*, and let* $\Phi' = \Phi + \mathcal{E}$ *with* $\mathcal{E}$ *gauge invariant of range at most* R*. Put* $t = \tanh\big(\frac{1}{2}\sum_{X\cap V\neq\emptyset} \|\mathcal{E}_X\|_\infty\big)$*. Then* Φ' *satisfies* $\mathrm{GDS}(V,\varepsilon')$ *with* $\varepsilon' = \varepsilon - 2t|V| \max_\nu |\partial V|_\nu / n_{\min}$*, where* $|\partial V|_\nu$ *is the number of links of direction* ν *in* ∂V*.*

Proof. The two energies on V differ by a function of oscillation at most $s = 2\sum_{X\cap V\neq\emptyset} \|\mathcal{E}_X\|_\infty$. Two probability measures whose log-densities with respect to a common measure differ by a function of oscillation s are within total variation $\tanh(s/4)$: if the density ratio takes values in $[c, ce^s]$, the total variation is at most $\sup_{0\le x\le 1}\big(\frac{xe^s}{1+x(e^s-1)} - x\big) = \tanh(s/4)$. Glue an optimal coupling of $\gamma'_V(\cdot \mid \omega)$ with $\gamma_V(\cdot \mid \omega)$, then $\Pi_{\omega,\omega'}$, then an optimal coupling of $\gamma_V(\cdot \mid \omega')$ with $\gamma'_V(\cdot \mid \omega')$. Then $\kappa'(y,b) \le \kappa(y,b) + 2t$, and summing over $b \in V$ and y of direction ν gives the stated loss. □

The smallness required in Lemma 6.6 shrinks with the box, since t must be of order $\varepsilon/(|V||\partial V|)$. This matters for Section 8.

6.4. **Two dimensions.** In two dimensions the criterion holds at every coupling. Take $G = \mathrm{SU}(2)$ and the plaquette weight $w(U) = e^{\frac{\beta}{2}\operatorname{Tr} U}$, write $c_0 = \int w$, $f = w/c_0$ and $a_j = I_{2j+1}(\beta)/I_1(\beta)$ for $j = \frac{1}{2}, 1, \dots$, where I_n is the modified Bessel function. The character expansion of f gives $f^{*m} = \sum_j (2j+1) a_j^m \chi_j$ and

$$\varepsilon_m := \sup_U |f^{*m}(U) - 1| = \sum_{j \geq 1/2} (2j+1)^2 a_j^m,$$

the supremum being attained at the identity because all a_j are positive. Since $0 < a_j < 1$ and $\sum_j (2j+1)^2 a_j < \infty$, $\varepsilon_m \to 0$ exponentially as $m \to \infty$.

Lemma 6.7 (Disk integration [20, 19])**.** *For a simply connected union D of k plaquettes, $\int \prod_{p \subset D} w(U_p)\, dU_{\mathrm{int}\, D} = c_0^k f^{*k}(\mathrm{hol}_{\partial D})$.*

Proof. Add the plaquettes one at a time in an order in which each new plaquette meets the region already built in a connected arc $u_1 \cdots u_r$. Substituting $g = u_1 \cdots u_r$ and using invariance of Haar measure gives $\int f^{*m}(Ag) f(g^{-1}B)\, dg = f^{*(m+1)}(AB)$, and the remaining arc variables integrate to 1. □

Proposition 6.8. *For every $\beta > 0$ there is L such that two-dimensional* SU(2) *Wilson theory satisfies* GDS(V, ε) *for the set V of the $2L(L-1)$ interior links of an $L \times L$ square and some $\varepsilon > 0$.*

Proof. For $y \in \partial V$ choose an $s \times s$ sub-square D with y on its outer side; let L_D be its $2s(s-1)$ interior links and $L_R = V \setminus L_D$. By Lemma 6.7, under $\gamma_V(\cdot \mid \omega)$ the marginal on L_R has density proportional to $\prod_{p \not\subset D} w(U_p) \cdot f^{*s^2}(\mathrm{hol}_{\partial D})$, and only the last factor depends on ω_y. Two densities proportional to ϱg_1 and ϱg_2 with $g_i \in [1-\epsilon, 1+\epsilon]$ are within total variation $\eta = 2\epsilon/(1-\epsilon^2)$. Couple the two marginals on L_R maximally and sample L_D independently; then $\kappa(y, b) \leq 1$ on L_D and $\kappa(y, b) \leq \eta_{s^2}$ on L_R, with $\eta_m = 2\varepsilon_m/(1-\varepsilon_m^2)$. Since $|\partial V|_\nu = 2L$ for each of the two directions and $n_{\min} = L(L-1)$, the condition (9) holds when $2L\big[2s(s-1) + 2L(L-1)\eta_{s^2}\big] < L(L-1)$, that is

$$\frac{4s(s-1)}{L-1} + 4L\,\eta_{s^2} < 1.$$

Choose s so large that $\eta_{s^2} < 1/(80s^2)$, which is possible because η_{s^2} decays exponentially in s^2, and then $L = 8s^2$. Both terms are then at most $\frac{1}{2} - \frac{1}{80}$. □

Remark 6.9. The consistency requirement mentioned in the introduction is the following. In four-dimensional U(1) theory at large β correlations decay only as a power [16, 15], so by Theorem 6.4 no box satisfies GDS there. A weaker hypothesis, bounding only the influence of one boundary link on one observable at the centre of the box, would be met by the massless spin-wave part of that theory for large boxes, because that influence decays like the fourth power of the box size; such a hypothesis cannot imply decay. The criterion has to measure joint couplings of the whole box, as Definition 6.3 does.

7. Transfer matrices and the volume

In this section $d = 4$ and $G \subset \mathrm{U}(N)$ is compact. Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$ be the spatial torus, with $3L^3$ links and $3L^3$ plaquettes, μ the Haar measure on G^{E_L} and $\mathcal{H}_L = L^2(G^{E_L}, \mu)$. In temporal

gauge the Wilson measure (3) on $\Lambda_L \times \mathbb{Z}/T\mathbb{Z}$ equals $\mathrm{Tr}(\mathbb{T}_L^T)/Z$ for the integral operator with kernel

$$k(U,V) = e^{\frac{N\beta}{2}\sum_p (\mathrm{Re\,Tr}\, U_p + \mathrm{Re\,Tr}\, V_p)} \int_{G^{\Lambda_L}} \prod_{e=(x,y)} e^{N\beta\, \mathrm{Re\,Tr}(U_e h_y V_e^{-1} h_x^{-1})} \prod_x dh_x,$$

the integral over h being the integral over the timelike links. This is the transfer matrix of Osterwalder and Seiler [22].

Proposition 7.1 (Finite-volume gap)**.** *For $\beta \geq 0$ the operator $\mathbb{T}_L$ is self-adjoint, nonnegative, Hilbert–Schmidt and positivity improving. Its largest eigenvalue λ_0 is simple with a strictly positive eigenfunction, and every other spectral value λ satisfies $|\lambda| \leq (1 - e^{-12N^2\beta L^3})\lambda_0$.*

Proof. The kernel is continuous on a compact space, hence Hilbert–Schmidt, and it is symmetric (substitute $h \mapsto h^{-1}$). Write $\mathbb{T}_L = A\mathcal{K}\Pi A$, with A multiplication by $e^{\frac{N\beta}{2}\sum_p \mathrm{Re\,Tr}\, U_p}$, $\mathcal{K}$ the convolution on each link by $\kappa(g) = e^{N\beta\,\mathrm{Re\,Tr}\, g}$ and Π the average over gauge transformations. Since $\mathrm{Re\,Tr}\, g = \frac{1}{2}(\chi_F(g) + \overline{\chi_F(g)})$ and products of characters decompose with nonnegative multiplicities, κ has nonnegative character coefficients, so convolution by it is a nonnegative operator; Π is an orthogonal projection commuting with $\mathcal{K}$, because κ is a class function, so $\mathcal{K}\Pi = \Pi\mathcal{K}\Pi \geq 0$ and $\mathbb{T}_L \geq 0$. From $|\mathrm{Re\,Tr}\, g| \leq N$ there are $3L^3 + 3L^3$ factors on each side of the bounds

$$m_L := e^{-6N^2\beta L^3} \leq k(U,V) \leq e^{6N^2\beta L^3} =: M_L,$$

so $\mathbb{T}_L$ is positivity improving. By Jentzsch's theorem [17] $\lambda_0 = \|\mathbb{T}_L\|$ is simple with an almost everywhere positive eigenfunction ψ_0, and $m_L\|\psi_0\|_1 \leq \lambda_0\psi_0 \leq M_L\|\psi_0\|_1$. The kernel $p(U,V) = k(U,V)\psi_0(V)/(\lambda_0\psi_0(U))$ is Markov, reversible for $\pi = \psi_0^2\mu/\|\psi_0\|_2^2$, and unitarily equivalent to $\mathbb{T}_L/\lambda_0$. It satisfies $p(U,V) \geq \epsilon\,\psi_0(V)/\|\psi_0\|_1$ with $\epsilon = m_L/M_L$, so by Doeblin's argument $\|P^n f\|_\infty \leq 2(1-\epsilon)^n\|f\|_\infty$ for bounded f with $\pi(f) = 0$. For the spectral measure σ_f of P at such f this gives $\int \lambda^{2n} d\sigma_f \leq 4(1-\epsilon)^{2n}\|f\|_\infty^2$ for all n, so σ_f vanishes on $\{|\lambda| > 1-\epsilon\}$. Bounded mean-zero functions are dense in the mean-zero subspace of $L^2(\pi)$, which proves the claim. □

This holds for U(1) as well and carries no information about infinite volume. Its role is to separate the problem.

Let ω be a state on the gauge-invariant local functions of $\mathbb{Z}^4$ that is invariant under translations and the hypercubic group and reflection positive for reflections in planes through sites and planes bisecting links. Every limit of Wilson states on periodic tori has these properties [22]. The reconstruction gives a Hilbert space with a vector Ω, a map $F \mapsto \Psi_F$ with dense range, and a self-adjoint $\mathbb{T}$ with $0 \leq \mathbb{T} \leq 1$, $\mathbb{T}\Omega = \Omega$ and $\langle \Psi_F, \mathbb{T}^t \Psi_H\rangle = \omega(\vartheta(\bar{F})\tau_t H)$, where ϑ is the time reflection and τ_t the time translation.

Proposition 7.2. *For $m > 0$ the following are equivalent:*

(a) *the spectrum of $\mathbb{T}$ on $\Omega^\perp$ lies in $[0, e^{-m}]$;*
(b) *for every bounded gauge-invariant F supported at times in $[0, s_F]$ there is $C_F < \infty$ with $0 \leq \omega(\vartheta(\bar{F})\tau_t F) - |\omega(F)|^2 \leq C_F e^{-mt}$ for all $t \geq 0$.*

Under (a) *one may take $C_F = \|F\|_\infty^2$.*

Proof. If (a) holds, $\omega(\vartheta(\bar{F})\tau_t F) - |\omega(F)|^2 = \langle \Psi'_F, \mathbb{T}^t \Psi'_F\rangle$ with $\Psi'_F = \Psi_F - \langle \Omega, \Psi_F\rangle\Omega \in \Omega^\perp$, which lies in $[0, e^{-mt}\|\Psi_F\|^2]$, and $\|\Psi_F\|^2 = \omega(\vartheta(\bar{F})F) \leq \|F\|_\infty^2$. Conversely, if σ is the spectral measure of $\mathbb{T}$ at Ψ'_F and $\delta > 0$, then $\sigma((e^{-m+\delta}, 1])e^{(-m+\delta)t} \leq C_F e^{-mt}$ for all t, so $\sigma((e^{-m+\delta}, 1]) = 0$; the spectral projection of $(e^{-m}, 1]$ restricted to $\Omega^\perp$ annihilates a dense set and vanishes. □

Let $\Delta_L(\beta) = \log(\lambda_0/\lambda_1)$ be the gap of $\mathbb{T}_L$ and ω_L its vacuum state.

Proposition 7.3. *If* $\inf_L \Delta_L(\beta) =: m > 0$*, every weak limit point* ω *of* (ω_L) *satisfies* $|\omega(F\tau_t H) - \omega(F)\omega(H)| \le \|F\|_\infty \|H\|_\infty e^{-mt}$ *for gauge-invariant* F, H *on single time slices, and hence has transfer-matrix gap at least* m*.*

Proof. The truncated correlation

$$\omega_L(F\tau_t H) - \omega_L(F)\omega_L(H) = \langle \bar{F}\Omega_L, (\mathbb{T}_L/\lambda_0)^t (1 - |\Omega_L\rangle\langle\Omega_L|) H\Omega_L\rangle$$

is bounded by $\|F\|_\infty \|H\|_\infty e^{-\Delta_L t}$. The bound is uniform in L and passes to weak limits, which exist by compactness. Proposition 7.2 completes the proof. □

Thus a positive gap on every finite torus is automatic for every compact group and every β, and a gap at fixed β in infinite volume is the uniformity of that gap in L. For four-dimensional U(1) at large β the uniformity fails, since the power-law decay of [15] and Proposition 7.2 give $\inf_L \Delta_L = 0$. Two structural facts separate SU(2) from U(1).

Proposition 7.4. *Call a map* $U_e \mapsto z_e U_e$ *with* $z\colon E \to Z(G)$ *and* $\prod_{e\in\partial p} z_e^{\pm 1} = 1$ *for every plaquette a centre transformation of the torus* $(\mathbb{Z}/L\mathbb{Z})^4$*. It preserves the Wilson weight, and modulo gauge transformations the centre transformations form* $H^1((\mathbb{Z}/L\mathbb{Z})^4; Z(G)) \cong Z(G)^4$*. For* $G = \mathrm{SU}(2)$ *this group is* $(\mathbb{Z}/2)^4$*; for* $G = \mathrm{U}(1)$ *it is the torus* $\mathrm{U}(1)^4$*.*

Proof. Central factors commute, so U_p is multiplied by $\prod_{e\in\partial p} z_e^{\pm 1} = 1$. Closed $Z(G)$-valued cochains modulo exact ones on the cubical complex of the torus form H^1 of the four-torus with coefficients in the abelian group $Z(G)$, which is $\mathrm{Hom}(\mathbb{Z}^4, Z(G)) = Z(G)^4$. □

Proposition 7.5. *Let* G *be semisimple and* F *a gauge-invariant* C^1 *function of finitely many links of* $\mathbb{Z}^d$*. At every flat configuration the differential of* F *in the directions of* $\mathfrak{g}$ *vanishes.*

Proof. On $\mathbb{Z}^d$ every flat configuration is a gauge transform of $U \equiv 1$, so we may take $U \equiv 1$, which is fixed by the constant gauge transformations $U_e \mapsto h U_e h^{-1}$. The differential $dF_1 \in (\mathfrak{g}^E)^*$ is invariant under the diagonal adjoint action, so its restriction to each factor is an Ad-invariant linear form on $\mathfrak{g}$. For semisimple $\mathfrak{g}$ the only such form is zero, because the adjoint representation has no invariant vectors. □

For U(1) the observable $\mathrm{Im}\, U_p$ has nonzero differential at $U \equiv 1$, and its correlations carry the massless photon. The massless photon of four-dimensional U(1) theory is the Goldstone mode of a continuous symmetry whose finite-volume shadow is the group $\mathrm{U}(1)^4$ of Proposition 7.4; for SU(2) that group is finite, and every local gauge-invariant observable is at least quadratic in the gauge field near flat configurations. Neither fact is an estimate, and neither excludes a critical point.

8. The missing step

The results above, and the cluster expansions they extend, control the Wilson theory below a fixed coupling. At the other end Bałaban's renormalisation group [3, 4] controls the block-spin flow (for the geometry of such flows in a gauge–Higgs setting see [7]), with bounds uniform in the lattice spacing, as long as the running coupling stays small. Between the two lies a window of order-one coupling crossed by a number of block-spin steps that does not depend on the spacing (Figure 4).

Problem 8.1 (Crossover)**.** Show that at a block scale $a_* = c_*/\Lambda$ the effective density of the four-dimensional SU(N) Wilson theory is a Wilson weight with coupling in the regime of Theorem B (or of a convergent cluster expansion) times $\exp \sum_X \mathcal{E}_a(X)$, with $\mathcal{E}_a$ gauge invariant,

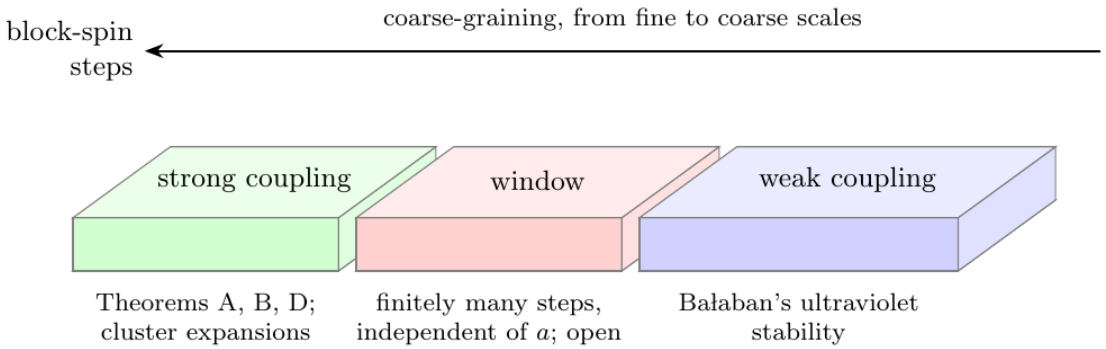


FIGURE 4. The three regimes of the block-spin flow. The strong-coupling end is controlled by the results of this paper and by cluster expansions, the weak-coupling end by Bałaban's ultraviolet stability. No method controls the window between them.

local, and small in the supremum norm over all block configurations, uniformly in the spacing a and in the volume.

A solution of Problem 8.1, followed by the strong-coupling analysis on the block lattice and reflection positivity of the fine theory, would give a transfer-matrix gap of at least ma at spacing a with m independent of a, which is the lattice form of the mass gap [6]; it would also give a gap at every fixed coupling, by Proposition 7.3. Problem 8.1 is open. The tools of this paper do not reach it: Theorem C and Proposition 5.3 show that the curvature and single-link Dobrushin methods stop at fixed small couplings because their extremal configurations (all plaquettes equal to z, or zero staple sum) are allowed, and Lemma 6.6 shows that feeding the corrections $\mathcal{E}_a$ into a finite-box criterion needs a smallness that shrinks with the box. The nearest test case, the crossover of the two-dimensional O(3) sigma model at weak coupling, is itself unsettled by refereed work, and three-dimensional SU(2) theory at large β is open as well.

Data and code availability

All proofs in this paper are by hand. The identities of Lemma 3.2, the extremal values in Theorem A and the Barta computations of Lemma 5.1 were also checked independently in exact arithmetic, and the bound of Theorem A against the numerically computed Hessian on random configurations; these checks are archived with the source of this paper in the repository `creelie/4Y-M`. The computations and checks were carried out by Deep Bhattacharjee in Python (SymPy, NumPy, SciPy) with AI assistance from Claude (Anthropic); both authors approved the final text.

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